

Test Plan and Report

Product Name: SlugRoute

Team Name: Team SlugRoute

Date: June 3, 2026

System Test Scenarios

The system has been evaluated against the following end-to-end user scenarios:

Scenario 1: Autocomplete and Course Selection (Pass)

1. Navigate to the application home page (<http://localhost:8080>) using Google Chrome.
2. Enter CSE 1 into the search bar.
3. **Verify output:** An autocomplete dropdown menu appears displaying matching courses (e.g., CSE 101, CSE 120).
4. Click on CSE 120 from the suggestion list.
5. **Verify output:** A search preview panel opens below the input bar, showing details for CSE 120 and checkboxes for associated discussion sections.
6. Select discussion section 01A and click the "Add to Map" button.
7. **Verify output:** The search bar clears, the preview dropdown closes, the course details card is appended to "Current Results" in the sidebar, and the map auto-zooms to focus on the plotted markers at Baskin Engineering.

Scenario 2: View Filter Controls (Pass)

1. Ensure CSE 120 (containing both lecture and discussion sections) is actively mapped on the screen.
2. In the "View Filters" section of the sidebar, uncheck the "Discussions" checkbox.
3. **Verify output:** The triangle marker at Baskin Engineering (representing the discussion section) disappears from the map instantly, while the star marker (representing the lecture) remains visible.
4. Re-check the "Discussions" checkbox.
5. **Verify output:** The triangle marker reappears on the map instantly.

Scenario 3: Walking Route Generation (Pass)

1. Click the "Set Start Point Manually" button (crosshair pin icon) in the map control stack.
2. Click on McHenry Library on the map layout.
3. **Verify output:** A blue circular "You Are Here" pin is drawn at McHenry Library.
4. Click on the CSE 120 lecture marker at Baskin Engineering.
5. **Verify output:** An InfoWindow appears displaying building details for "J Baskin Engr".
6. Click the blue arrow button in the upper right of the InfoWindow.

7. **Verify output:** A blue walking polyline is rendered on the map, running from McHenry Library to Baskin Engineering. A floating stat bubble appears near the destination, indicating the estimated walking time (e.g., "7 min") and distance.

Scenario 4: Schedule Export to iCalendar (Pass)

1. Verify at least one course is visible in the results sidebar.
2. Click the "Sync to Calendar" button in the sidebar footer.
3. **Verify output:** The button text shifts to "Generating File..." and the browser initiates a download for a file named `slugroute-schedule-2026.ics`.
4. Open the downloaded `.ics` file in a text editor.
5. **Verify output:** The file structure begins with `BEGIN:VCALENDAR`, includes a valid `VTIMEZONE` block for `America/Los_Angeles`, and details the scheduled events containing correct names, locations, and recurrence rules (`RRULE:FREQ=WEEKLY`).

Automated Unit Tests

Automated testing is integrated into our release branch across multiple layers of the application stack. All automated tests compile and pass cleanly:

1. Backend Go Tests

- **Directory:** `backend/`
- **Execution Command:** `cd backend && go test ./...`
- **Test Coverage:** Evaluates API routing logic, term fetching, autocomplete queries, database integrations, and handling of optional data fields in `main_test.go`.
- **Result:** Pass

2. Python Scraper Tests

- **Directory:** `scraper/`
- **Execution Command:** `python3 scraper/test_scraper.py`
- **Test Coverage:** Validates PISA page extraction parsing, text normalization, and date-to-STRM calculation helpers.
- **Result:** Pass

3. Database Migration Tests

- **Directory:** `database/`
- **Execution Command:** `python3 database/test_import_coords.py`
- **Test Coverage:** Verifies coordinate reading, uppercase name normalization, and error handling for missing coordinates or images.
- **Result:** Pass

4. Frontend Javascript Tests

- **Directory:** `frontend/`
- **Execution Command:** Start the backend server (`go run .` in `/backend`), then open `http://localhost:8080/tests` (or open the local `frontend/tests.html` file) in a web browser.
- **Test Coverage:** Evaluates browser mocking, course format utilities, state clustering logic, and marker prioritizations.
- **Result: Pass**